PROJECT REPORT

MEDICAL DIAGNOSIS CENTER BY SERVICE PORTAL (SERVICENOW)

A ServiceNow-based solution for managing diagnostic center services

Project Details	Information
Student Name	Karra Deepak
Department	Computer Science and Engineering – Artificial Intelligence and Machine Learning
Institution	Lendi Institute of Engineering and Technology, Jonnada, Vizianagaram
Project Platform	ServiceNow Service Portal
Academic Year	2026–2027

1. INTRODUCTION

1.1 Project Overview

Medical Diagnosis Center by Service Portal (ServiceNow) is a centralized digital platform developed to simplify diagnostic center operations. It enables patients to register, explore available tests, book appointments, track booking status, and view laboratory reports. Built on ServiceNow, the system helps reception staff, laboratory personnel, and administrators manage patient information, test details, appointments, and reports efficiently.

he main objective is to reduce manual work, improve information accessibility, and establish a smooth workflow between patients and diagnostic center staff.

1.2 Purpose

This study focuses on developing a Medical Diagnosis Center Management System on the ServiceNow platform to organize and streamline diagnostic center activities. The application brings essential services, including patient registration, test booking, appointment management, sample handling, and laboratory report access, together within a single Service Portal.

The proposed system is intended to minimize paperwork and provide patients with a convenient way to schedule tests, monitor appointment progress, and view their reports. It also enables diagnostic center staff to maintain patient information, update test progress, and handle reports systematically. ServiceNow supports centralized data management and role-based access for patients, receptionists, laboratory technicians, and administrators.

The overall goal is to improve the accessibility of diagnostic services, simplify administrative tasks, maintain well-organized records, and support the efficient management of daily diagnostic center operations.

2. IDEATION PHASE

2.1 Problem Statement

Diagnostic centers often face difficulties in managing patient registration, diagnostic test bookings, appointments, sample processing, and laboratory reports through manual methods or separate record-keeping systems. These processes can require considerable time and effort from staff and may lead to errors, misplaced records, delays, and difficulties in tracking the progress of diagnostic tests.

Patients may experience inconvenience when booking diagnostic tests, checking appointment details, and obtaining laboratory reports. They may need to contact the diagnostic center repeatedly to know whether their samples have been processed or their reports are available. Maintaining patient information, appointment details, and test results separately can also make it difficult for staff to retrieve accurate records and manage the complete diagnostic workflow efficiently.

Furthermore, the absence of a centralized system can create communication gaps between reception staff, laboratory technicians, and patients. Staff may find it challenging to maintain organized records, update test statuses, and provide patients with timely access to their diagnostic information.

To address these challenges, the proposed project, "Medical Diagnosis Center by Service Portal ServiceNow," aims to develop a centralized and user-friendly web-based system using the ServiceNow platform. The system is designed to simplify patient registration, provide diagnostic test booking facilities, organize appointment records, support laboratory staff in updating sample processing statuses, and enable patients to access their laboratory reports through a single Service Portal.

The proposed system focuses on reducing manual effort, improving record management, simplifying communication, and making diagnostic services more convenient for patients and staff while maintaining organized and secure access to diagnostic information.

2.2 Empathy Map Canvas

Empathy Area	Patient / Customer Perspective
Says	"I want to book a diagnostic test without repeated visits or calls."
Thinks	"I need to know when my appointment is scheduled and when my report will be ready."
Does	Visits or contacts the center, selects tests, attends appointments, and asks for report updates.
Feels	May feel uncertain or inconvenienced when booking details and report status are not readily available.
Pain Points	Manual booking, repeated follow-ups, unclear test status, and difficulty locating reports.

Needs / Gains	Simple test selection, appointment confirmation, progress tracking, and convenient report access.

2.3 Brainstorming

- Provide a single portal for diagnostic test information and patient services.
- Allow patients to register and submit diagnostic test appointment requests.
- Maintain appointment and diagnostic test details as ServiceNow records.
- Enable authorized laboratory staff to update sample and test progress.
- Allow patients to view or download available laboratory reports securely.
- Provide administrators with record-management capabilities and available reporting or dashboard tools.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Patient Action	System / Staff Support	Expected Outcome
Discover	Opens the Service Portal and browses diagnostic tests.	Displays available test details.	Patient identifies a required test.
Register	Provides registration and profile information.	Stores patient details in the system.	Patient record is available.
Book	Selects a test and submits an appointment request.	Creates and tracks an appointment record.	Booking request is recorded.
Attend / Process	Visits the center for the scheduled test.	Authorized staff update sample and test status.	Progress is traceable.
View Report	Opens My Reports when the report is available.	Provides access to the associated report.	Patient can view or download the report.

3.2 Solution Requirement

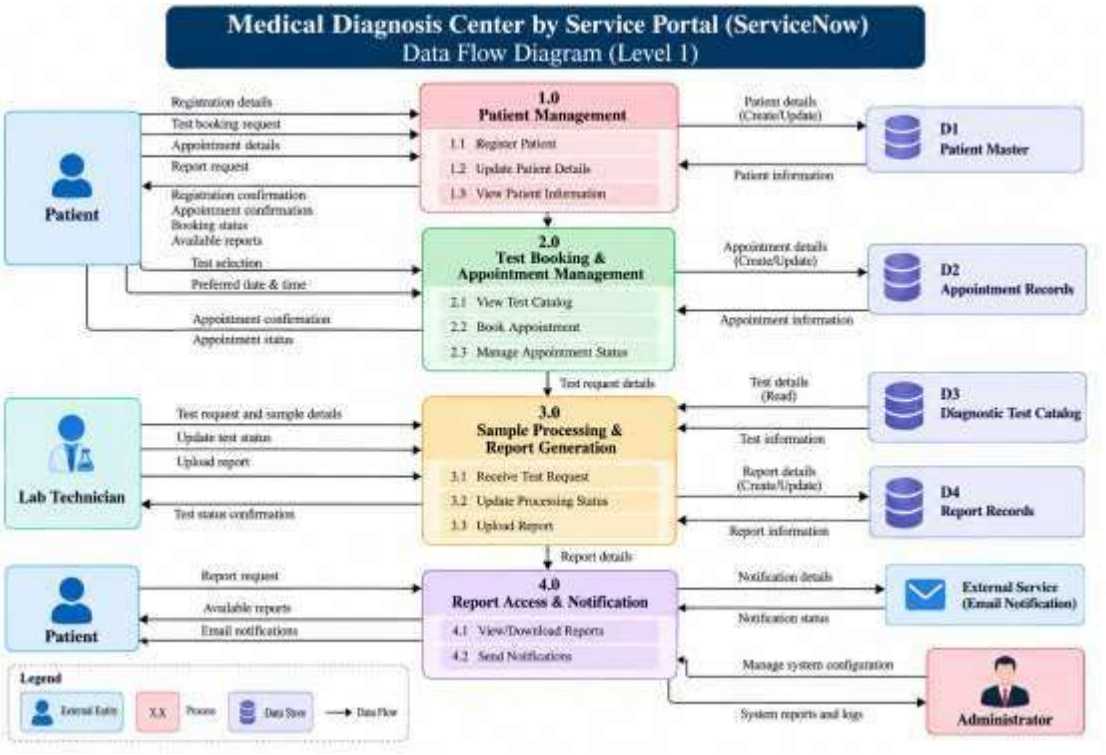
Category	Requirements
Functional	Patient registration; diagnostic test catalog; appointment booking; booking history; sample/test status updates; report viewing or downloading; administrative management of tests and records.
User Roles	Patient, reception staff, laboratory staff, and administrator. Access should be restricted according to assigned permissions.
Data	Patient details, diagnostic test information, appointment records, sample/test status, and laboratory reports or attachments.
Platform	ServiceNow platform and Service Portal; platform tables, forms, portal components, and workflow/business-rule capabilities as configured.
Security	Use authenticated access and appropriate role/record access controls. Patient reports should be available only to authorized users.

3.3 Data Flow Diagram

Level 0 – Context description: The Patient interacts with the Service Portal to register, browse tests, book appointments, track status, and access reports. Reception staff manage booking-related records. Laboratory staff update sample and test progress and attach or maintain reports. The Administrator maintains test information and oversees records.

External Entity	Data Sent to System	Data Received from System
Patient	Registration details, selected test, appointment request	Booking details, status, available report
Reception Staff	Appointment updates and patient service information	Patient and appointment records
Laboratory Staff	Sample status, test status, report information	Test and appointment details
Administrator	Test catalog and administrative updates	Operational records and available summaries

Level 1 – Main processes: (1) Manage Patient Registration, (2) Manage Diagnostic Test Catalog, (3) Process Appointment Booking, (4) Update Sample/Test Status, and (5) Manage and Provide Laboratory Reports. The processes read and update the corresponding ServiceNow records and related report attachments.



3.4 Technology Stack

Technology / Area	Project Use
ServiceNow Platform	Core platform for records, forms, access control, and workflow configuration.
Service Portal	Patient-facing interface for diagnostic services.
HTML	Structure of portal interface components where applicable.
CSS	Styling and layout of portal pages and widgets.
JavaScript / ServiceNow APIs	Client-side interactions and platform scripting, where configured.
ServiceNow Tables	Storage and management of patient, test, and appointment data.
Attachments / Reports	Storage or access to laboratory report documents, subject to permissions.
Business Rules / Flow Designer	Automation and record updates where implemented.

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem Identified	Proposed Fit
Manual booking and reception dependency	Portal-based diagnostic test selection and appointment request submission.
Scattered patient, test, and appointment information	Related ServiceNow records support centralized tracking.
Repeated calls for progress updates	Status information can be updated by authorized staff and checked through the portal.
Difficulty accessing reports	Secure report viewing or downloading when a report is available.

4.2 Proposed Solution

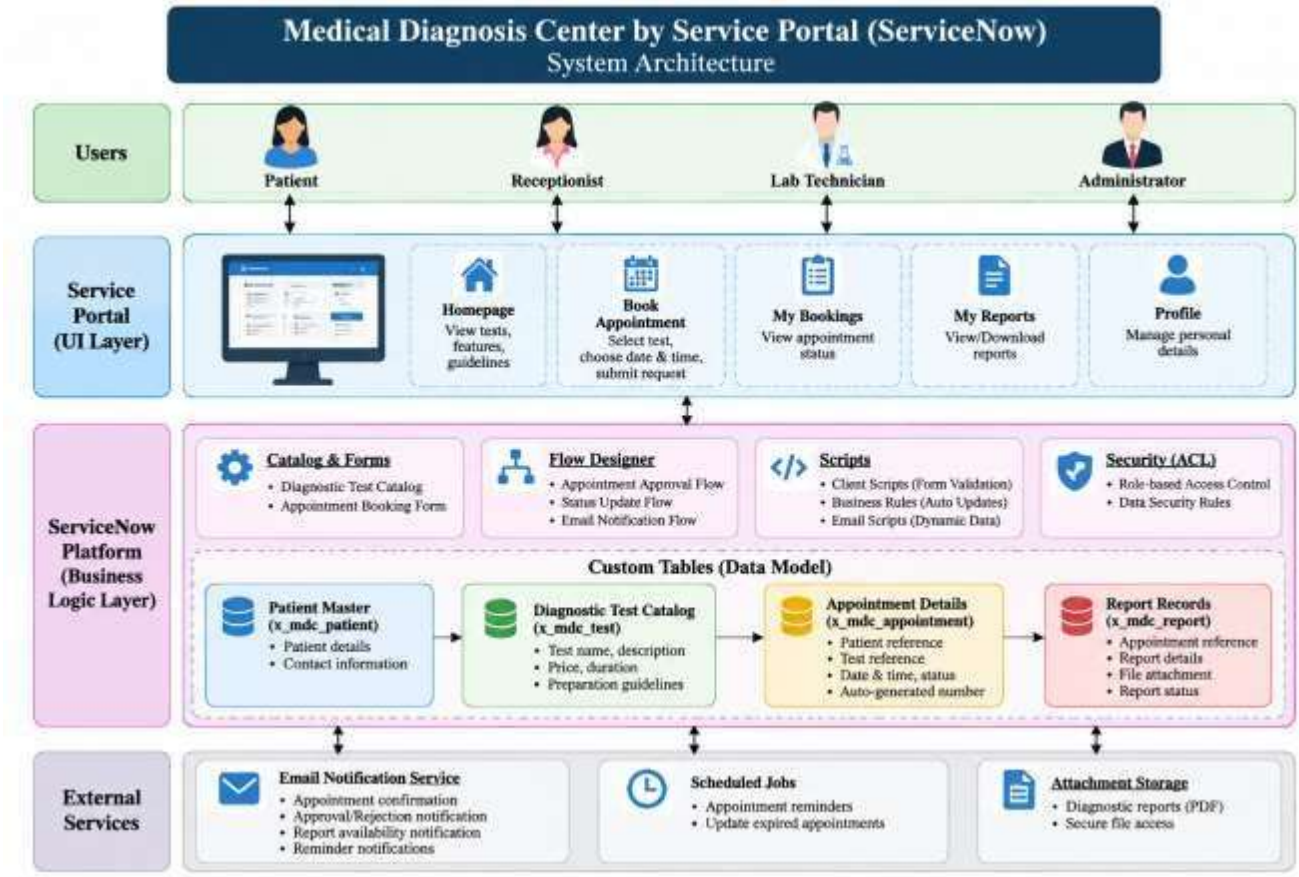
The proposed solution is a ServiceNow-based diagnostic service portal that connects patient-facing activities with staff record management. Patients can register, browse the available diagnostic tests, and submit appointment requests. Reception staff can manage booking information, while laboratory staff update sample and test statuses and maintain report information. Patients can then check progress and access reports through the portal, subject to configured access permissions.

Administrators can maintain diagnostic test information and oversee the records and available reports. The project is intended to improve organization, reduce avoidable manual coordination, and provide a more convenient diagnostic service experience.

4.3 Solution Architecture

Layer	Components / Responsibility
Presentation Layer	Service Portal pages and widgets for patient registration, test catalog, bookings, status tracking, and reports.
Application / Logic Layer	ServiceNow scripts, configured business rules, flows, and form logic to process requests and update records.
Data Layer	ServiceNow tables for patient, diagnostic test, and appointment information; report attachments or related report records.
User / Access Layer	Patient, reception staff, laboratory staff, and administrator roles with appropriate access controls.

Workflow: Patient or staff action → Service Portal / ServiceNow form → validation and configured logic → ServiceNow record update → status or report made available to authorized users.

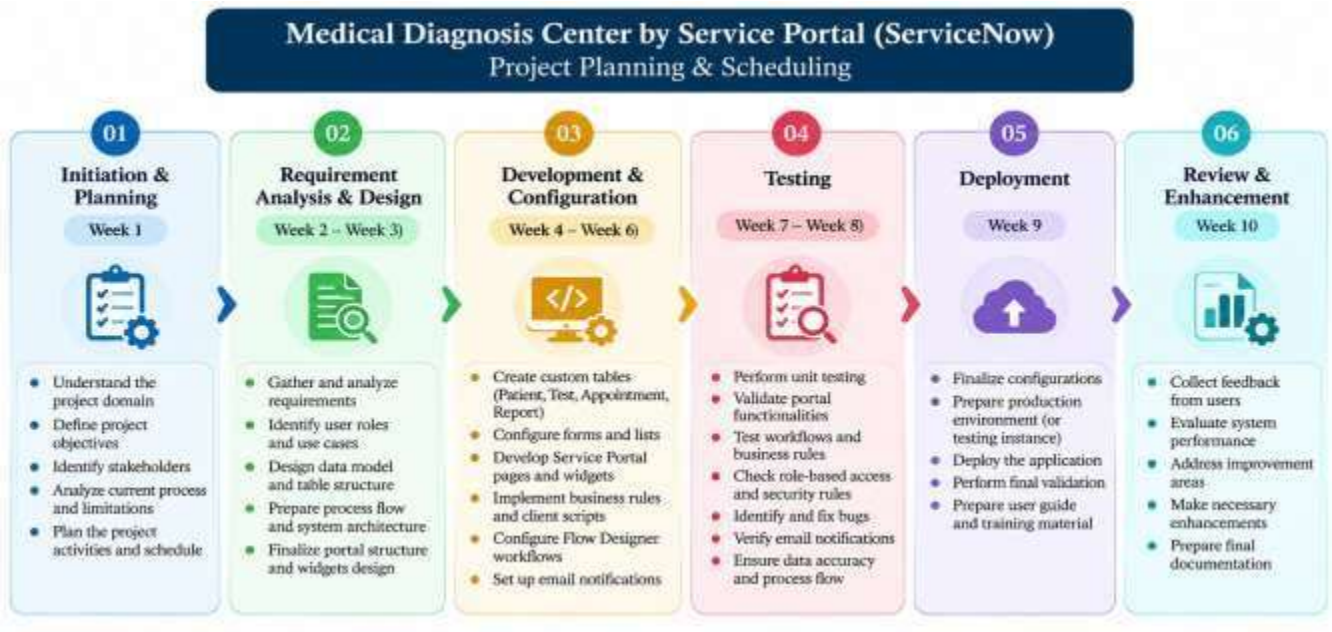


5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase / Sprint	Tasks	Deliverable
Sprint 1 – Requirements	Identify users, problems, requirements, and diagnostic workflow.	Problem statement and requirement list.
Sprint 2 – Data and Design	Define tables, fields, roles, portal structure, and process flow.	Data design and architecture.
Sprint 3 – Portal Development	Create portal pages, test catalog, registration, and booking features.	Patient-facing portal features.
Sprint 4 – Workflow and Reports	Configure status updates, staff processes, and report access.	Diagnostic workflow and report features.
Sprint 5 – Testing and Documentation	Run functional checks, capture output screenshots, and prepare report.	Test evidence and final documentation.

The sprint plan is a project-planning outline. Actual dates, assigned team members, and completion status should be updated according to the team's work log.



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing should verify that core portal features work as intended and that the time required for common actions is recorded. The results below are a test plan and must be updated with actual measured values from the ServiceNow instance.

Test ID	Test Scenario	Expected Result	Actual Result / Status
TC-01	Open diagnostic test catalog.	Available tests and details are displayed.	To be measured / verified
TC-02	Submit a valid appointment request.	Appointment record is created and confirmation is displayed.	To be verified
TC-03	Open My Bookings.	Relevant appointment records are displayed to the authorized patient.	To be measured / verified
TC-04	Update sample or test status as lab staff.	Authorized status update is saved and visible as configured.	To be verified
TC-05	Open an available laboratory report.	Authorized user can view or download the report.	To be verified
TC-06	Trigger configured email notification.	Notification is processed and delivered to the configured recipient.	To be verified

7. RESULTS

7.1 Output Screenshots

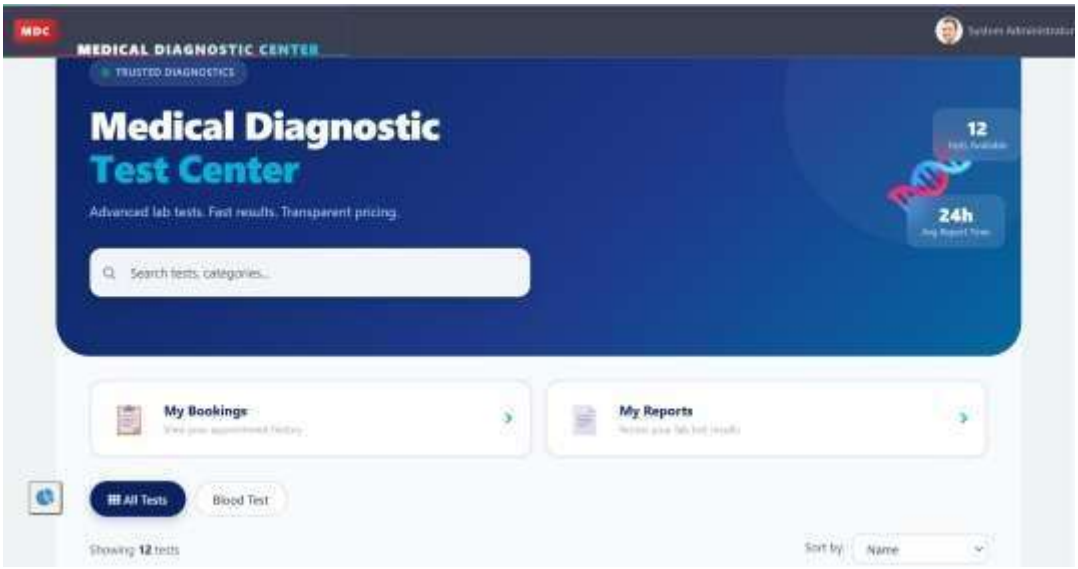


Figure 1: Medical Diagnosis Center Home Page

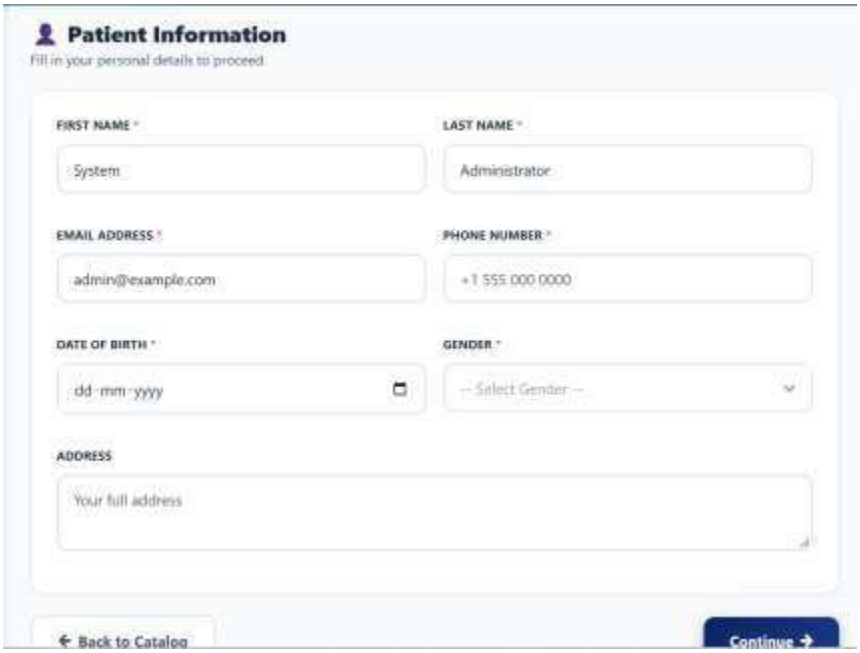


Figure 2: Patient Registration Form

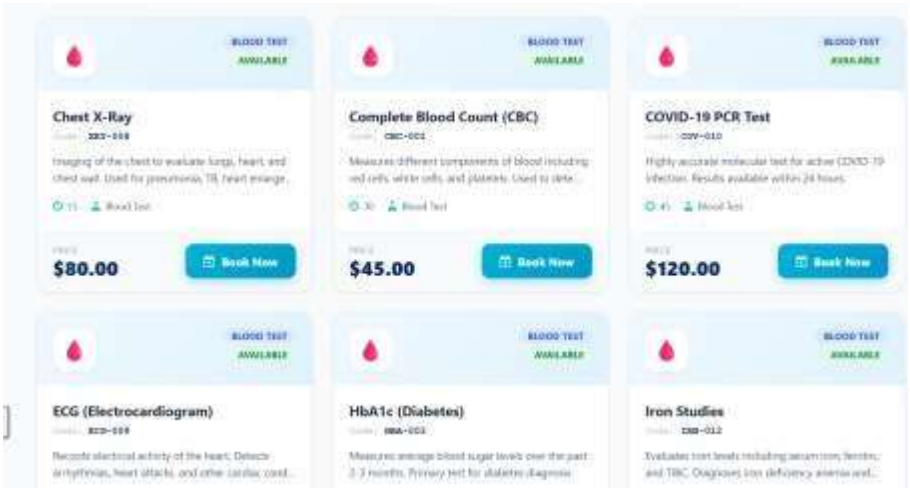


Figure 3: Diagnostic Test Catalog

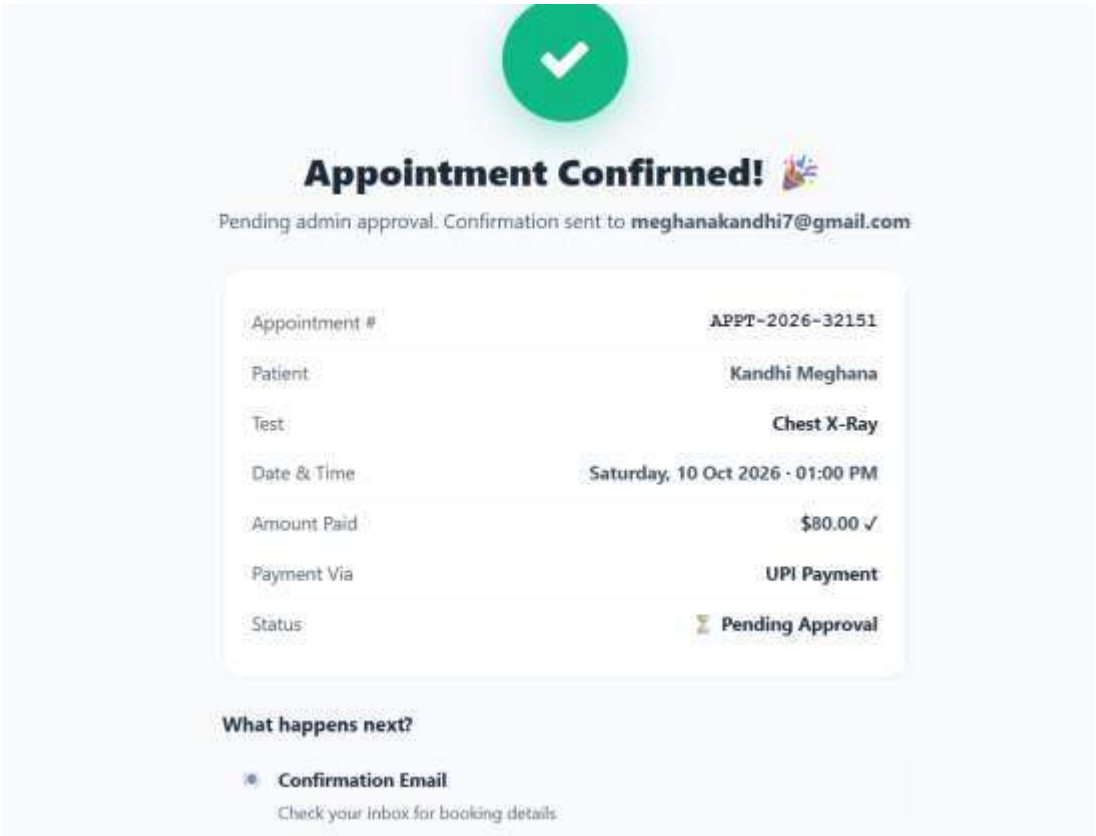


Figure 4: Diagnostic Appointment Confirmation

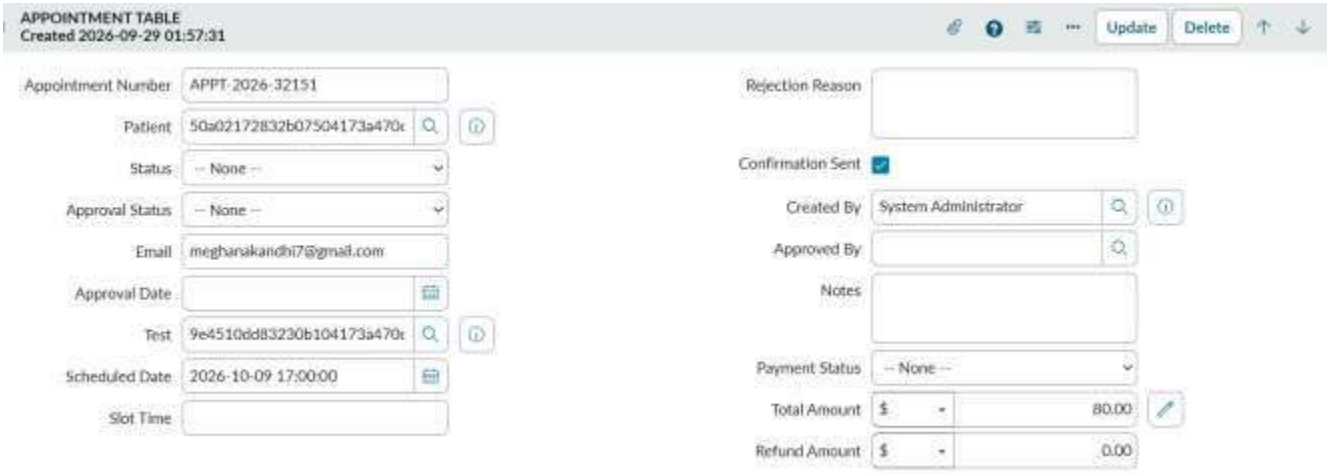


Figure 5: Appointment Record Management in ServiceNow

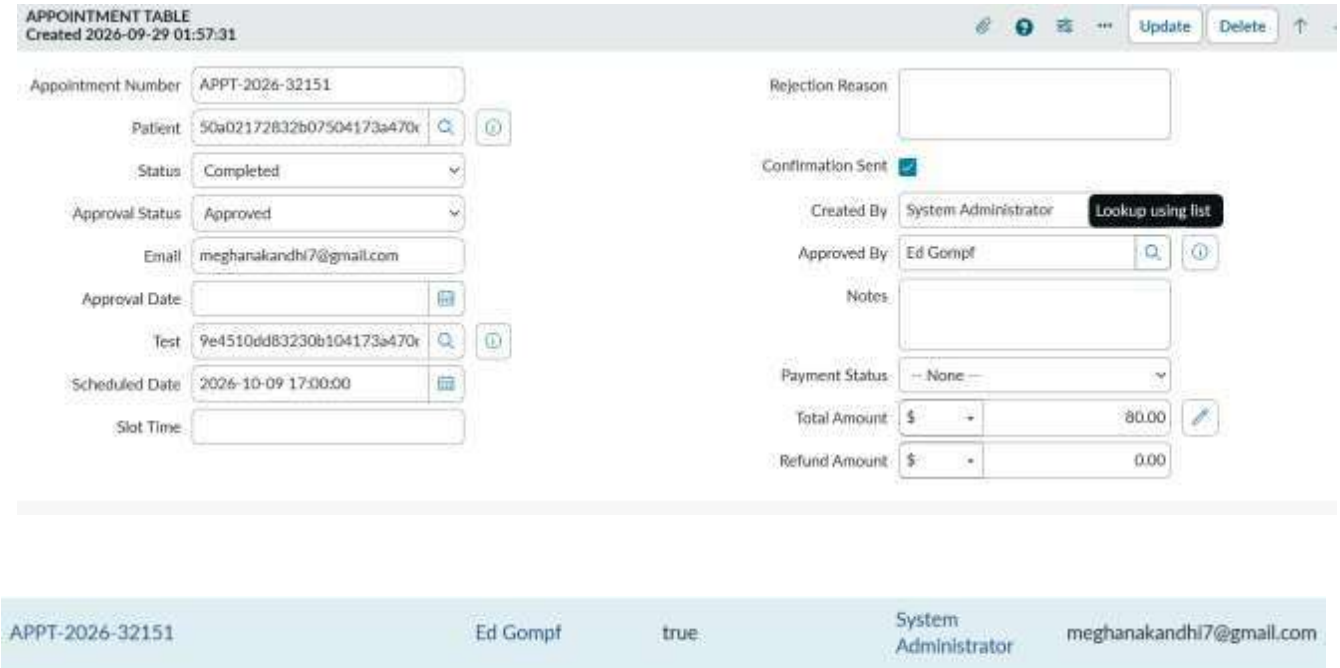


Figure 6: Appointment Confirmation Record

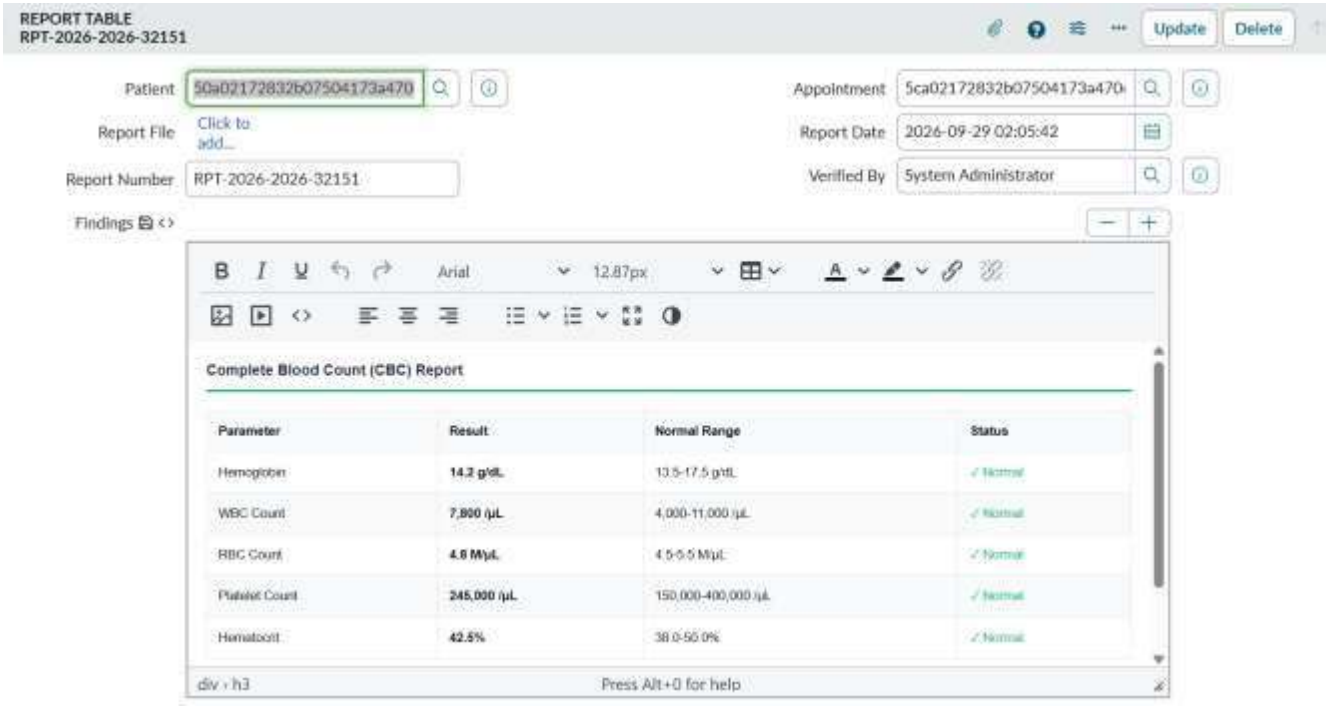


Figure 7: Laboratory Report Management in ServiceNow

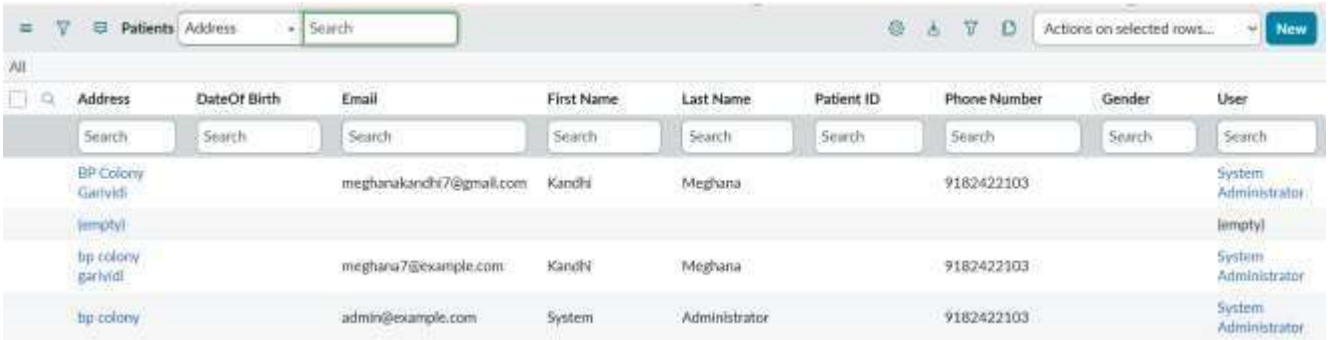


Figure 8: Patient Registration Records in ServiceNow

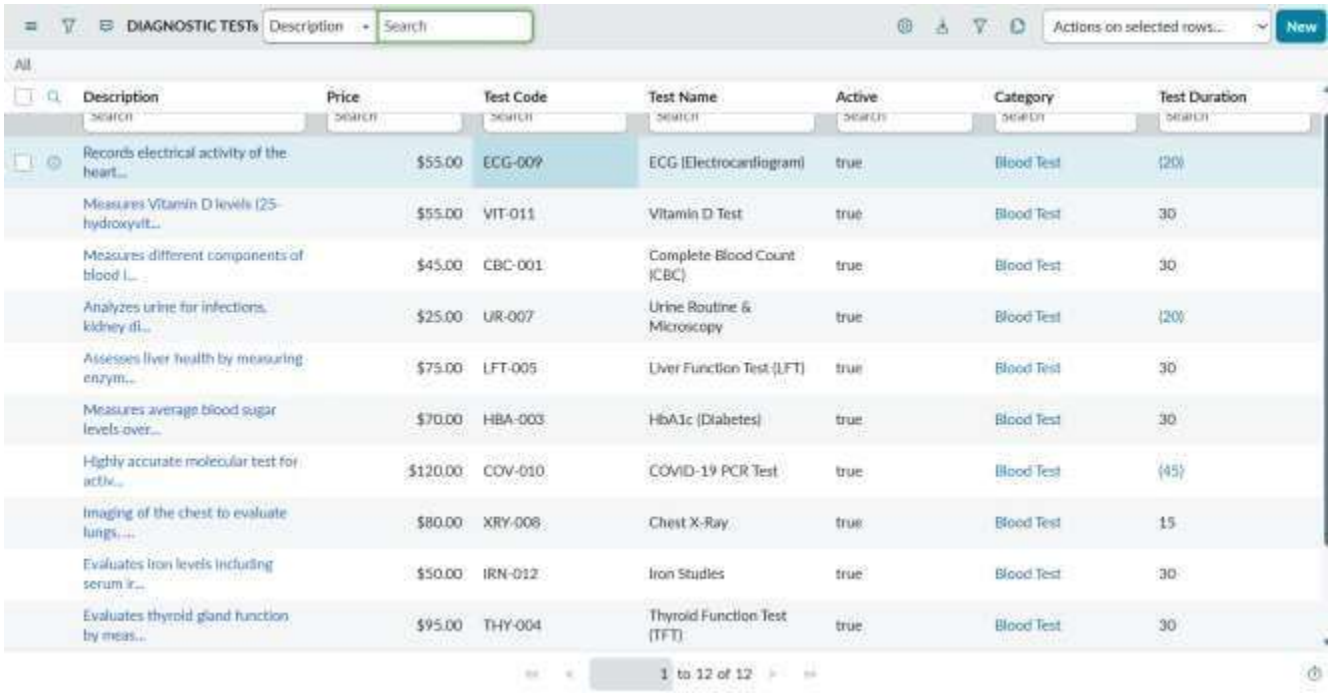


Figure 9: Diagnostic Test Catalog Management in ServiceNow



Figure 10: Appointment Approval Email Notification in ServiceNow

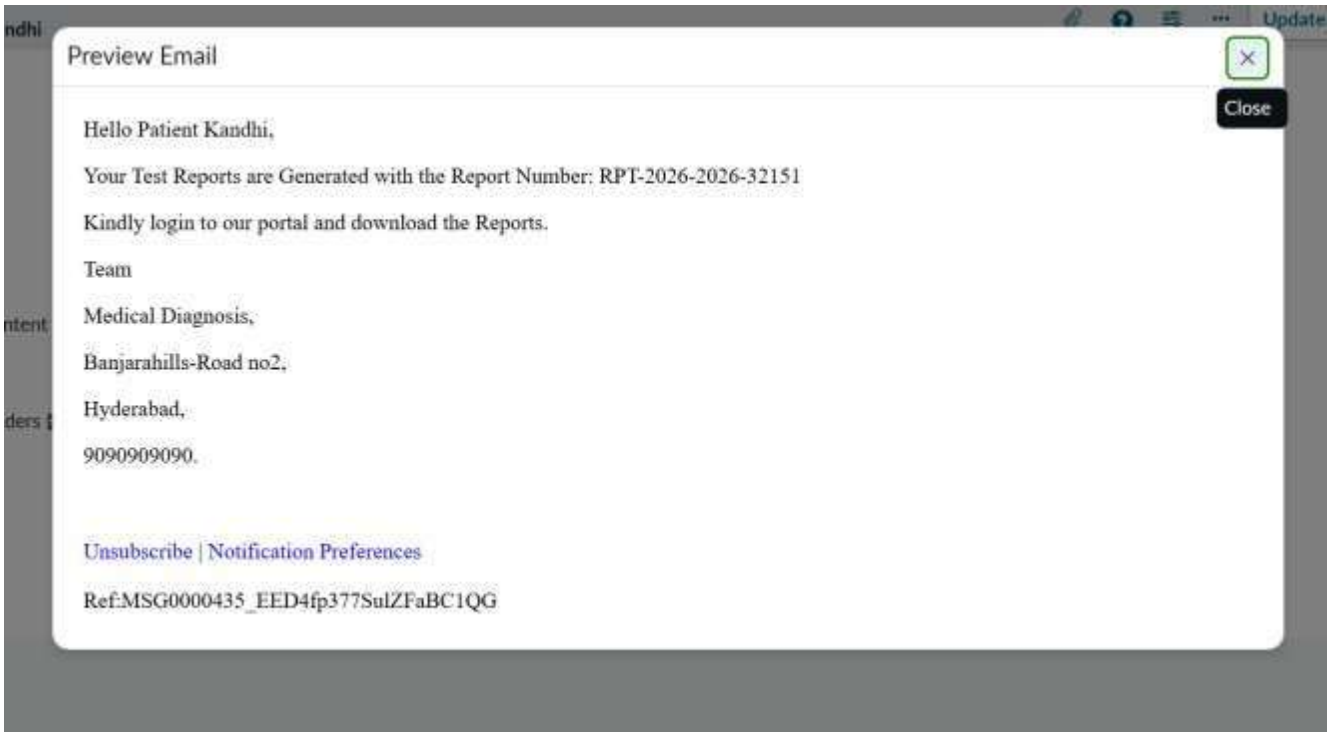


Figure 11: Laboratory Report Generation Email Notification

8. ADVANTAGES & DISADVANTAGES

1. Advantages

1. Centralized Management: The system provides a centralized Service Portal for managing patient registration, diagnostic test bookings, appointments, and laboratory reports. It helps maintain information in an organized manner and makes records easier to access and manage.

2. Reduced Manual Work: The system reduces the need for repeated manual booking activities and frequent status inquiries. Patients can use the portal to book diagnostic tests and check their appointment details, helping reduce the workload of diagnostic center staff.

3. Improved Appointment and Test Tracking: The system helps staff maintain appointment records, update test statuses, and track the progress of diagnostic services. This makes it easier to manage booking information and follow the diagnostic process.

4. Convenient Report Access: Patients can access available laboratory reports through the Service Portal. This reduces the need to depend entirely on staff for report retrieval and provides a convenient way to view or download diagnostic results.

5. ServiceNow Platform Support: The project uses ServiceNow features such as custom tables, forms, Service Portal, and workflow configuration to manage diagnostic center activities. These capabilities support organized record management and provide a foundation for further system development.

2. Disadvantages / Limitations

1. Dependence on System Configuration: The proper functioning of the system depends on the correct configuration of ServiceNow roles, access controls, and workflows. Incorrect configuration may affect record access, user permissions, and the functioning of different processes.

2. Email Delivery Dependency: Email notifications depend on the outbound email configuration of the ServiceNow instance. Although notification templates can be configured, successful delivery depends on the email settings and mail server. Therefore, email delivery must be verified separately.

3. No External Laboratory Equipment Integration: The current project does not establish real-time integration with external laboratory equipment or diagnostic machines. Sample processing updates and report information depend on the system's configured processes and staff updates.

4. Performance and Network Dependency: The performance of the application may vary depending on ServiceNow instance configuration, network connectivity, and the amount of stored data. The system's response time and performance under different workloads need to be measured through testing.

5. Limited Medical Functionality: The system is designed to manage diagnostic bookings, records, test statuses, and laboratory reports. It does not replace doctors, medical professionals, or professional interpretation of diagnostic results. Patients should consult qualified healthcare professionals for medical advice and diagnosis.

9. CONCLUSION

The project “Medical Diagnosis Center by Service Portal (ServiceNow)” demonstrates the use of the ServiceNow platform to organize and simplify the daily activities of a medical diagnostic center. The system provides a centralized Service Portal for managing important activities such as patient registration, diagnostic test selection, appointment booking, appointment status tracking, and laboratory report access.

The project aims to reduce the difficulties associated with manual record maintenance and separate management of diagnostic services. By maintaining patient details, appointment information, diagnostic test records, and laboratory reports in a centralized system, it helps organize information and makes it easier for authorized users to access and manage records.

The system also provides separate workflows for patients and diagnostic center staff according to their responsibilities. Patients can use the portal to view available diagnostic tests, book appointments, track their booking details, and access available laboratory reports. Staff can manage patient records, maintain appointment information, update diagnostic test statuses, and manage laboratory report details. This structured approach aims to improve coordination between patients and staff and make diagnostic services more convenient.

The use of ServiceNow features such as Service Portal, custom tables, forms, and workflow configuration demonstrates how a service management platform can be applied to healthcare administration. The project provides a foundation for digitizing diagnostic center operations and improving the organization and traceability of diagnostic records.

Although the system supports the main diagnostic management activities, further testing and verification are necessary to evaluate its overall reliability and effectiveness. In particular, email notification delivery, application performance, and access control functionality should be tested and measured to strengthen the final evaluation of the project.

In conclusion, the Medical Diagnosis Center by Service Portal provides a centralized and structured approach to diagnostic center management. It aims to reduce manual effort, improve record organization, and provide convenient access to diagnostic services for patients and staff. The project can be further enhanced through additional features, improved performance, and integration with external laboratory systems in the future.

10. FUTURE SCOPE

The Medical Diagnosis Center by Service Portal can be further enhanced by adding new features and technologies to improve diagnostic center operations, patient convenience, and system efficiency.

1. Online Payment Integration

In the future, the system can be integrated with secure online payment gateways to allow patients to pay diagnostic test fees directly through the Service Portal. This can simplify the payment process and help maintain payment transaction records along with appointment details.

2. SMS and Email Notifications

The system can be enhanced with improved notification services to send appointment confirmations, reminders, approval updates, and report availability alerts through SMS and email. This can help patients stay informed about their bookings and reduce the need for repeated inquiries.

3. Integration with Laboratory Equipment

The system can be integrated with external laboratory equipment and diagnostic machines to receive test results automatically. This can reduce manual data entry, improve the consistency of report information, and support faster updates to diagnostic records.

4. Advanced Appointment Scheduling

Future development can include an advanced appointment scheduling system with available time slots, laboratory workload management, and automatic slot allocation. This can help prevent booking conflicts and support better management of patient appointments.

5. Mobile Application Development

A dedicated mobile application can be developed to provide patients with convenient access to diagnostic services through smartphones. Patients could register, book tests, track appointments, receive notifications, and view laboratory reports from their mobile devices.

6. Enhanced Security and Data Protection

The system can be improved by implementing stronger authentication methods, additional access control rules, and enhanced data protection mechanisms. These improvements can help protect sensitive patient information and ensure that diagnostic records are accessible only to authorized users.

7. AI-Based Diagnostic Assistance

In the future, artificial intelligence can be explored to provide general health information and assist in organizing diagnostic data. Such features could help patients understand basic information about laboratory tests and reports. Any medical interpretation or diagnostic assistance would require clinical validation and professional medical supervision.

8. Dashboard and Analytical Reports

The system can be enhanced with advanced dashboards and analytical reports to display information such as the number of appointments, frequently booked tests, pending reports, and completed diagnostic services. These features can help administrators understand operational activities and support better planning.

9. Multi-Branch Diagnostic Center Management

The system can be extended to support multiple diagnostic center branches through a centralized platform. Patients could select a preferred branch, while administrators could manage branch-wise appointments, diagnostic test catalogs, and operational records.

11. APPENDIX

GitHub & Project Demo Link

GitHub: https://github.com/Meghanakandhi/Medical-Diagnosis-Center-by-Service-Portal-ServiceNow-.git

Project Demo: [Add your demo URL or recording link here]

Before submission, verify that the repository and demo links are accessible to the intended reviewers and do not expose credentials, patient information, or other confidential data.